

Instructor Guide

Python Programming Course — Lab Setup Session

Companion to the Student Lab Setup Guide

Purpose of This Guide

This guide supports the instructor or lab assistant running the first-class environment setup session described in the Student Lab Setup Guide. It lists the issues that most commonly come up on Windows laptops while students install Python, VS Code, Git, and connect to GitHub, along with practical fixes, plus things to watch for while circulating the room. It assumes the same numbering of parts (Part 1 through Part 18) used in the student handout.

Tip: Keep the Student Lab Setup Guide open next to this one — the part numbers below match it directly, so you can jump straight to the relevant step if a student calls you over.

Before-Class Preparation

A few minutes of preparation before students arrive prevents most of the issues below.

- Confirm every lab laptop (or student laptop, if BYOD) runs Windows 10 or 11 and has at least ~2 GB of free disk space and administrator rights for the current user, or know in advance which machines are locked down.
- Download the offline installers to a USB drive as a backup in case the room Wi-Fi cannot handle 20–30 simultaneous downloads: python-3.13.15-amd64.exe (from the Files table on the 3.13.15 release page), the VS Code User Installer, and Git for Windows/x64 Setup (2.55.0 at the time of writing).
- Keep that exact Python installer for the whole course. Python 3.13 is scheduled to move to security-only releases around October 2026, and python.org does not publish Windows installers for security-only releases — later 3.13 versions may exist only as source code.
- Check that github.com, python.org, code.visualstudio.com, and the VS Code Marketplace are not blocked by the institution's firewall or content filter.
- If laptops are institution-managed, confirm with IT beforehand whether Python, VS Code, and Git are already deployed, already blocked, or need a help-desk ticket to install.
- Check whether any lab machines already have Anaconda, Miniconda, or an older Python version installed — these are the most common source of PATH conflicts (see Part 2 below).
- Have one or two laptops set up ahead of time as a known-good reference so you can quickly compare against a student's screen.
- Remind students in advance to bring a laptop charger — installs, especially Git and VS Code, can take a while on battery-saving settings.

Suggested Time Allocation

For a class of 20–30 students with mixed prior experience, a rough budget for a single session:

Time	Parts covered	Notes
10 min	Part 1 — GitHub account	Fastest part; watch for email verification delays.
20 min	Parts 2–4 — Python, VS Code, Python extension	Slowest part; installers and restarts take time.
15 min	Parts 5–8 — Git install/config, course folder	Mostly quick once Python/VS Code are done.
20 min	Parts 9–13 — venv, interpreter, first program	Most error-prone section; budget extra time here.
10 min	Part 14 — pip check	Quick, but confirm venv is active first.
15 min	Parts 15–17 — GitHub connect, Git commands, .gitignore	May slip to a second session if time is short.
5 min	Part 18 — Final verification checklist	Walk the room and sign off students individually.

Tip: If the session is only 60 minutes, it is reasonable to stop after Part 14 and cover Parts 15–17 (GitHub connection) at the start of the next lab, once everyone has a working local Python setup.

Common Issues by Part

These are grouped in the same order as the student guide so you can find them quickly.

Part 1 — GitHub Account

Symptom: Verification email never arrives, or arrives 10+ minutes late.

Likely cause: Institutional email filters often quarantine GitHub's verification email, especially on school or college domains.

Fix: Have the student check Spam/Junk first. If it still has not arrived after a few minutes, let them continue with the rest of the setup and verify the email later — nothing else in the guide requires a verified account immediately.

Symptom: Student already has a GitHub account from a previous course or personal project.

Likely cause: Not actually a problem — the guide allows reusing an existing account.

Fix: Confirm they can sign in and remember the password; no new account is needed.

Watch for: Some students will try to sign up with a school email that requires SSO or is not a normal inbox. Recommend a personal email address if the school email causes friction.

Part 2 — Install Python 3.13

Symptom: "python --version" opens the Microsoft Store, or prints nothing.

Likely cause: Windows ships "app execution alias" stub files for python.exe and python3.exe that redirect to the Store when no real Python is on PATH, or when the alias takes priority over the real install.

Fix: Go to Settings → Apps → Advanced app settings → App execution aliases, and turn off the python.exe / python3.exe entries. Then open a new Command Prompt and try again.

Symptom: "python --version" shows Python 2.7, or a much older 3.x version, or an Anaconda build.

Likely cause: A previous Python or Anaconda installation is earlier on PATH than the new Python 3.13 install.

Fix: Either uninstall the older distribution if it is not needed, or, at minimum, make sure students select the Python 3.13 interpreter explicitly in VS Code (Part 11) rather than relying on the plain python command.

Symptom: Installer never prompted for "Add python.exe to PATH", or the student unchecked it by mistake.

Likely cause: The checkbox on the first installer screen was missed.

Fix: Simplest fix: re-run the Python installer, choose "Uninstall", then reinstall from scratch with the box checked. Manually editing the PATH environment variable also works but is more error-prone for a first lab session.

Symptom: "Windows protected your PC" (SmartScreen) warning when running the installer.

Likely cause: The installer is unsigned or not yet recognized by SmartScreen's reputation database — this is expected behaviour for a freshly downloaded installer from python.org.

Fix: Confirm the file was downloaded from the official python.org link in the guide, then click "More info" → "Run anyway".

Symptom: python --version shows Python 3.14.x, or the student installed the "Python install manager" instead of Python 3.13.

Likely cause: The large Download Python button on the python.org home page now gives 3.14, and the 3.13.15 release page shows a Download Python install manager link above the installer list.

Fix: Uninstall the 3.14 entry (Settings → Apps → Installed apps), then run python-3.13.15-amd64.exe from the Files table on the 3.13.15 page, with Add python.exe to PATH selected. The student guide now warns about both links (Part 2, Step 1).

Symptom: Installer will not run, or asks for an administrator password the student does not have.

Likely cause: The laptop is institution-managed and the student account lacks install permissions.

Fix: Use the "Install for me only" (per-user) option in the Python installer, which does not require administrator rights. If that option is also blocked, escalate to IT — this is a device policy issue, not a setup mistake.

Part 4 — Python Extension in VS Code

Symptom: Extensions panel shows "Unable to search the Marketplace" or install spins forever.

Likely cause: A campus proxy or firewall is blocking the VS Code Marketplace domain.

Fix: Check with IT whether marketplace.visualstudio.com needs to be allow-listed. As a workaround, the extension can be installed later once network access is confirmed; it is not required to reach Part 12.

Watch for: Searching "Python" in the Extensions panel returns several look-alike extensions. Walk the room once during this step and glance at each screen to confirm the installed one shows publisher "Microsoft".

Parts 5–6 — Install and Configure Git

Symptom: "git is not recognized as an internal or external command" right after installing.

Likely cause: The terminal or Command Prompt window was opened before Git finished installing, so it has not picked up the updated PATH.

Fix: Close all open terminals and Command Prompt windows, then open a brand-new one.

Symptom: Student is unsure which options to pick during the Git installer.

Likely cause: The Git for Windows installer has many screens (line-ending handling, default editor, credential helper, etc.).

Fix: Tell students to accept the defaults on every screen for this course; the defaults work correctly with the rest of the guide.

Watch for: In Part 6, suggest the private GitHub address (Settings → Emails → Keep my email addresses private, ...@users.noreply.github.com) for git config user.email, so students' personal email addresses do not appear in public commit history.

Parts 9–10 — Virtual Environment (venv)

Symptom: Activating .venv in the VS Code terminal gives: "cannot be loaded because running scripts is disabled on this system."

Likely cause: The default VS Code terminal on Windows is PowerShell, which blocks running local scripts (including activate.ps1) under its default execution policy.

Fix: The student guide now has students switch the VS Code terminal to Command Prompt before activating (Part 9, Step 2: terminal ▾ menu → Select Default Profile → Command Prompt, then open a new terminal). If a student is still in PowerShell, either switch profile as above or run Set-ExecutionPolicy -Scope CurrentUser RemoteSigned once, answer Y, and retry. Both fixes were tested on Windows 10 with the default Restricted policy.

Symptom: Creating the virtual environment is extremely slow, or VS Code reports files are locked/in use.

Likely cause: The PythonCourse folder lives inside a OneDrive-synced Documents folder, and OneDrive is actively trying to sync the thousands of small files venv creates.

Fix: Either pause OneDrive syncing while working in this folder, or move the PythonCourse folder to a path outside OneDrive (for example, directly under C:\Users\YourName\PythonCourse) for the course.

Symptom: Virtual environment creation fails with a path-related error.

Likely cause: The full folder path is very long (common with deeply nested OneDrive or school-managed profile paths), hitting Windows' path-length limits.

Fix: Move the project folder closer to the drive root to shorten the path, for example C:\PythonCourse instead of a deeply nested Documents path.

Part 11 — Select the Interpreter

Symptom: The .venv interpreter does not appear in the "Python: Select Interpreter" list.

Likely cause: VS Code has not refreshed its interpreter list since the .venv folder was created.

Fix: Wait a few seconds and reopen the Command Palette, or choose "Enter interpreter path" and browse to .venv\Scripts\python.exe directly inside the project folder.

Parts 12–14 — First Program and pip

Symptom: Running hello.py shows old output, no output, or errors referencing a different Python version.

Likely cause: The interpreter shown in the VS Code status bar is not the .venv one — often because the student skipped Part 11 or picked the wrong entry.

Fix: Check the bottom-right status bar in VS Code; it should read something like Python 3.13.x ('.venv': venv). Reselect the correct interpreter if not.

Parts 15–17 — Connecting to GitHub

Symptom: git clone or git push prompts for a username and password, and the student's normal GitHub password is rejected.

Likely cause: GitHub no longer accepts account passwords for Git operations over HTTPS; a personal access token or the bundled Git Credential Manager sign-in flow is required instead.

Fix: Let the sign-in popup from Git Credential Manager (installed with Git for Windows by default) complete in the browser, and approve it there. If a token is asked for directly, the student needs to generate a personal access token from GitHub's Developer Settings — this is worth demonstrating once to the whole class rather than repeating it one-on-one.

Symptom: git clone fails with a connection or timeout error.

Likely cause: The lab network blocks outbound Git traffic to github.com, or the repository URL was mistyped.

Fix: Confirm the exact URL from the instructor, and check with IT whether github.com and its SSH/HTTPS ports are reachable from the lab network.

Things to Watch For While Students Are Working

- Circulate continuously rather than waiting at the front of the room — most of the issues above are visible at a glance (wrong interpreter in the status bar, missing (.venv) prefix, SmartScreen dialog sitting unanswered).
- Do not let a struggling student silently fall behind. If one student's issue is taking more than a couple of minutes, ask them to keep working on the current step (or move to the next one) while you note the issue and return, rather than holding up the room.
- Watch for students installing extra software they think they need (Anaconda, a different code editor, an unrelated "Python" extension) — redirect them back to the guide.
- Warn the room in advance not to close the terminal or Command Prompt window mid-installation, and not to unplug or let the laptop sleep during installers.
- Keep an eye out for students copying commands from chat apps or messaging tools, which can silently substitute curly quotes for straight quotes in commands like git config --global user.name "Name" and cause confusing errors.
- For early finishers, have them help a neighbor or start Part 15 (connecting to the course repository) rather than sitting idle.
- Do a visible pass at Part 18 rather than trusting a show of hands — ask each student to show you the (.venv) prompt and the Hello, Python! output before they leave.

Quick-Reference Troubleshooting Table

A condensed version of the issues above, for a fast lookup while walking the room.

Symptom	Likely Cause	Fix
python --version opens Microsoft Store	App execution alias intercepting python.exe	Disable in Settings → Apps → Advanced app settings → App execution aliases
Wrong Python version shown	Old Python/Anaconda earlier on PATH	Select .venv interpreter explicitly in VS Code (Part 11)
"Add to PATH" was missed	Checkbox skipped during install	Uninstall and reinstall Python with the box checked
SmartScreen warning on installers	Unsigned/new installer, expected behaviour	"More info" → "Run anyway", confirm official source
Installer needs admin rights	Locked-down/managed laptop	Use per-user install option, or escalate to IT
Marketplace search fails in VS Code	Network/proxy blocking Marketplace	Check firewall allow-list with IT; retry later
"git" not recognized	Terminal opened before PATH refreshed	Close and reopen a new terminal window
PowerShell blocks activate.ps1	Default execution policy restricts scripts	Switch VS Code terminal to Command Prompt (Part 9, Step 2), or Set-ExecutionPolicy -Scope CurrentUser RemoteSigned
venv creation very slow / files locked	OneDrive syncing the project folder	Pause OneDrive sync or move folder outside OneDrive
venv creation fails on path error	Path too long (deep OneDrive/profile nesting)	Move project folder closer to drive root
.venv not listed as an interpreter	VS Code has not refreshed interpreter list	Reopen Command Palette, or enter the path manually
Program output looks wrong/old	Wrong interpreter selected in status bar	Reselect the .venv interpreter (Part 11)
git push/clone rejects password	GitHub no longer accepts passwords over HTTPS	Use Git Credential Manager sign-in or a personal access token
git clone times out	Network blocks github.com, or URL mistyped	Verify URL; check GitHub access with IT

When to Escalate to IT Rather Than Troubleshoot Live

A few situations are worth recognizing quickly as device or network policy issues rather than setup mistakes, since they cannot be fixed from within the student guide:

- The student's account cannot install any software, even with the per-user Python installer option.
- python.org, code.visualstudio.com, or github.com are unreachable from the lab network entirely (not just slow).
- The VS Code Marketplace is blocked and no offline extension package (.vsix) is available.
- Antivirus software silently deletes or quarantines the downloaded installers.

Tip: For these cases, keep the affected student moving through whatever parts of the guide do not depend on the blocked resource, and follow up once IT has resolved the underlying policy or network issue.

After the Session

- Note which students did not complete the Part 18 checklist and follow up individually before the next lab.
- Record any recurring issue that affected multiple students (for example, a shared lab-image problem or a blocked domain) and report it to IT so it does not repeat in the next session.
- If Parts 15–17 were postponed due to time, confirm at the start of the next session that every student can still activate their `.venv` and run `hello.py` before moving on to GitHub.

Guide v1.1 — updated 26 Sep 2026 to match Student Lab Setup Guide v1.1 (Python 3.13.15, VS Code 1.139.1, Git for Windows 2.55.0).